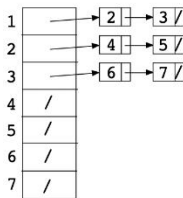
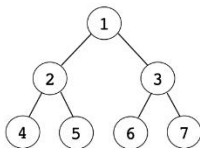


Question 1

Give an adjacency-list representation for a complete binary tree on 7 vertices. Give an equivalent adjacency-matrix representation. Assume that vertices are numbered from 1 to 7 as in a binary heap. (Edges are directed from parent to child)



Question 2

You are given a graph represented as adjacency lists. Perform Breadth First Search (BFS) starting from the node *s*. Implement the traversal using a queue, and demonstrate step-by-step how the algorithm works, including the state of the queue, visited nodes, and the final traversal order.

$$\text{adj}(s) = [a, c, d],$$

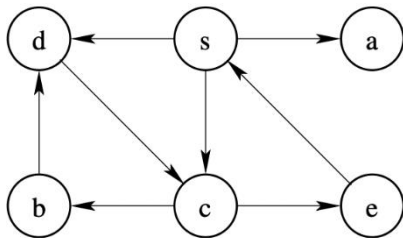
$$\text{adj}(a) = [],$$

$$\text{adj}(c) = [e, b],$$

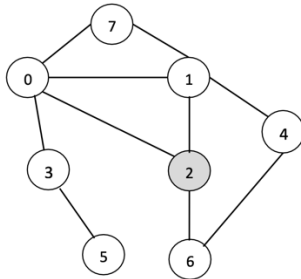
$$\text{adj}(b) = [d],$$

$$\text{adj}(d) = [c],$$

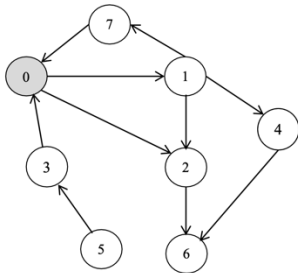
$$\text{adj}(e) = [s].$$



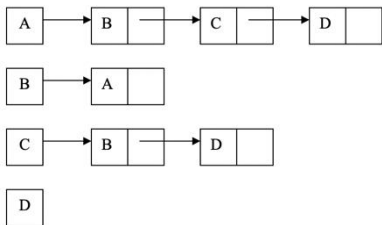
Question 3: Traverse the graph below in Breadth-First order, starting from vertex 2. Whenever you examine neighbors of a vertex, process them in increasing order of vertex number. Half the points will be lost if you visit the neighbors in a different order.



Question 4: Traverse this graph with Depth First Search (DFS) (all DFS, not just one DFS-Visit). If a vertex has more than one neighbors you must visit them in increasing order of vertex number, for example from node 1 you would visit the neighbors in order: 2,4,7. Half the points will be lost if you visit the neighbors in a different order.



Question 5: Here is an adjacency list representation of a directed graph where there are no weights assigned to the edges).



(a) Draw a picture of the directed graph that has the above adjacency list representation.

(b) Another way to represent a graph is an adjacency matrix. Draw the adjacency matrix for this graph.

Question 6: Build the Huffman coding tree for the message:

Huffman trees are fun

1. Calculate the frequency of each character in the message (including spaces).
2. Construct the Huffman coding tree step by step.
3. Assign binary codes to each character using the tree.
4. Encode the entire message using the derived codes.

Note: Show all steps, including the intermediate stages of tree construction, and provide the final encoded message.